MVP

MVP Document: ShopForMe

Project Title: ShopForMe

Date: May 21, 2024

Version: 1.1

---

1. Project Overview

Objective: Enhance the shopping experience for the blind or visually impaired by providing a mobile application that communicates with IoT devices to assist users in navigating stores.

Problem Statement: Navigating a store can be challenging for blind or visually impaired individuals due to the difficulty in locating specific items.

Proposed Solution: A mobile application that uses voice control to manage shopping lists and provides audio feedback with precise directions to item locations using IoT technology.

Expected Benefits: Greater independence for visually impaired shoppers and a more accessible shopping environment.

---

2. MVP Objectives

- **Primary Objectives:**
  - Implement voice-controlled list management and product search.
  - Provide accurate real-time navigation inside stores.
  - Deliver audio feedback for directions and aisle confirmation.
- **Success Criteria:**
  - Precision in location and direction information.
  - User satisfaction with the voice interaction model.
  - Effective audio feedback that enhances user navigation.

---

3. Personas and Use Cases

Target Users:

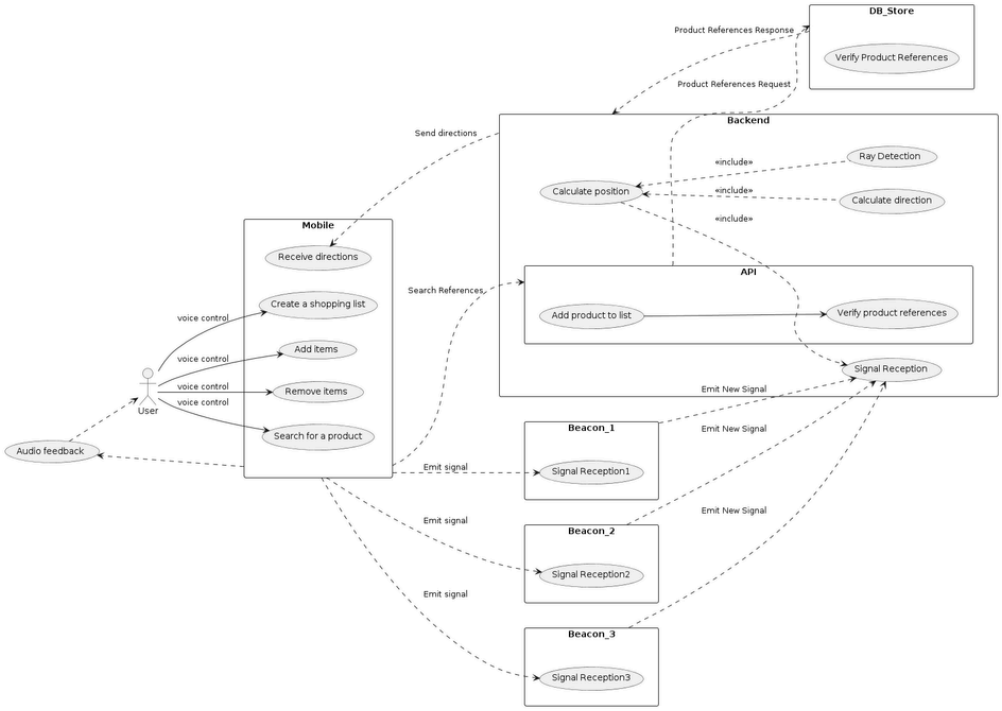
- **Persona 1:** John Doe, visually impaired, frequent smartphone user.
- **Persona 2:** Jane Smith, blind, active technology user for accessibility.

Use Case:

- **Scenario:** Jane Smith wants to find pasta in the supermarket. She uses the app to add "pasta" to her shopping list with voice commands. The app, interacting with in-store IoT devices, guides her directly to the pasta aisle with audio directions.

Use Case:

- Code schéma




```
@startuml
left to right direction
actor User

rectangle Mobile {
  User --> (Create a shopping list) : voice control
  User --> (Add items) : voice control
  User --> (Remove items) : voice control
  User --> (Search for a product) : voice control
  (Receive directions)
}

rectangle Beacon_1 {
  Mobile ..> (Signal Reception1) : Emit signal
}

rectangle Beacon_2 {
  Mobile ..> (Signal Reception2) : Emit signal
}

rectangle Beacon_3 {
  Mobile ..> (Signal Reception3) : Emit signal
}

rectangle Backend {
  Beacon_1  ..> (Signal Reception) : Emit New Signal
  Beacon_2  ..> (Signal Reception) : Emit New Signal
  Beacon_3  ..> (Signal Reception) : Emit New Signal
  (Calculate position) ..> (Signal Reception) : <<include>>
  (Calculate position) <.. (Calculate direction) : <<include>>
  (Calculate position) <.. (Ray Detection) : <<include>>
```

```

  rectangle API {
    (Add product to list) --> (Verify product references)
  }
}

rectangle DB_Store {
  (Verify Product References)
}

Mobile ..> API : Search References
API ..> DB_Store : Product References Request
DB_Store ..> Backend : Product References Response

Backend ..> (Receive directions) : Send directions
Mobile ..> (Audio feedback)
(Audio feedback) ..> User

@enduml
```

4. Key Features

- **Voice Control**: Users can interact with the app using voice commands to manage their shopping list.
- **IoT Integration**: IoT devices in the store receive signals from the mobile app to help locate the user.
- **Real-Time Navigation**: Backend algorithms calculate the user's position and send navigational directions to the mobile app.
- **Audio Feedback**: Directions and aisle confirmations are provided audibly to assist users in finding items.

---

5. Technical Specifications

- **Technologies Used:**
  - **IoT Devices:** Equipped with Bluetooth Low Energy.
  - **Mobile Application:** Built with Flutter for cross-platform compatibility.
  - **Backend:** ?
- **Data Flow:**
  - The user interacts with the mobile app through voice commands.
  - The mobile app sends signals to IoT devices.
  - IoT devices relay location data to the backend.
  - The backend processes this data, calculates directions, and sends them back to the mobile app.
  - The user receives these directions as audio feedback.

---

6. System Architecture

This section refers to the provided UML diagram, detailing each component's role and interaction.

---

7. Project Team

- **Project Manager:** @Frederic LE DOUARIN
- **Mobile Developer:** @Rémi Perray, @Corentin Ribeyron
- **IoT Developer: @Franques Matthieu, @Maxime Delmon, @Elhoucine Haddouch**
- **Backend Developer:** @Franques Matthieu, **@Maxime Delmon, @Elhoucine Haddouch**
- **QA Engineer:** ?

---

8. Risks and Constraints

- **Potential Risks:**
  - Delays in signal transmission could affect real-time response.
  - Challenges in achieving high accuracy in positioning within certain store layouts.
- **Constraints:**
  - Dependency on the store's infrastructure for IoT device placement.
  - Budget constraints for extensive IoT setup.

---

9. Glossary

- **IoT:** Internet of Things, referring to the network of physical devices that gather and communicate data.
- **BLE:** Bluetooth Low Energy, used for short-range communication between devices.
- **Flutter:** A framework for building native apps using React